

Supplementary Appendix for: Ψ -Risk-DT: A Neurosymbolic Framework for Adaptive Zero-Day Threat Detection in Digital Twin Ecosystems

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OVERVIEW

This appendix consolidates and extends the proposed improvements to the mathematical formalism of “ Ψ -Risk-DT: A Neurosymbolic Framework for Adaptive Zero-Day Threat Detection in Digital Twin Ecosystems”. The refinements are organized thematically and aim to enhance consistency, reproducibility, and theoretical soundness.

A. Hybrid Loss Function

The hybrid loss is defined as

$$L_{\text{total}} = \alpha L_{\text{cls}} + \beta L_{\text{graph}} + \gamma L_{\text{sem}}, \quad \alpha + \beta + \gamma = 1, \quad (\text{A.1})$$

where the normalization constraint ensures interpretability and stability. Hyperparameters can be determined through Bayesian optimization or similar automated search strategies. The reference matrix W_{gt} is defined as

$$W_{\text{gt}} = \mathcal{M}(G_t), \quad (\text{A.2})$$

where $\mathcal{M}$ maps the RDF graph G_t into a weighted adjacency matrix, with semantic predicates (e.g., `:hasEntropySpike=1`) used as weights.

The semantic distance is specified as the squared Euclidean norm,

$$L_{\text{sem}} = \sum_{(s,p,o) \in T_t} \|\text{emb}(a_t) - \text{emb}((s,p,o))\|_2^2, \quad (\text{A.3})$$

which is differentiable and Lipschitz-continuous, with constant bounded by $2 \max \|\text{emb}\|$. Embeddings are formally defined as

$$\text{emb} : (s, p, o) \mapsto \mathbb{R}^{128}, \quad (\text{A.4})$$

obtained via RDF2Vec with random walks of length three.

Entropy modulation is handled through normalization of ΔH_t :

$$\Delta H_t = \frac{H(X_t) - H_{\text{base}}}{\max(\varepsilon, H_{\text{base}})} \in [0, 1], \quad (\text{A.5})$$

with the modulation parameter bounded as $\lambda \leq 0.2$ in

$$\hat{L}_{\text{total}} = L_{\text{total}} \cdot (1 + \lambda \Delta H_t). \quad (\text{A.6})$$

The discrepancy between activations and RDF triples is given by

$$\Delta(a_t, G_t) = \frac{1}{|T_t|} \sum_{(s,p,o) \in T_t} \|a_t - \text{emb}((s,p,o))\|_2, \quad (\text{A.7})$$

while graph coherence is defined as

$$\text{Coh}(G_t) = 1 - \frac{\# \text{ inconsistent triples}}{\# \text{ total triples}}. \quad (\text{A.8})$$

The Lyapunov-like functional,

$$V(\vartheta) = \alpha \|y - \hat{y}\|^2 + \beta \|W_t - W_{\text{gt}}\|_F^2 + \gamma (1 - \text{Coh}(G_t))^2, \quad (\text{A.9})$$

decreases over iterations provided that the learning rate satisfies $\eta < 2/L$, where L is the Lipschitz constant of the Hessian. While L_{graph} is convex, the terms L_{cls} and L_{sem} are not; bounding their subgradients is thus necessary to guarantee stability.

B. Neurosymbolic Operator Ψ

The neurosymbolic operator is expressed as

$$\Psi : (\Delta H_t, x_t, G_t) \mapsto (a_t, G_{t+1}). \quad (\text{A.10})$$

It is essential to clarify whether ΔH_t is a scalar, representing joint entropy deviation, or a vector, capturing per-feature entropies. Norms and thresholds must be chosen consistently with this representation. The activation function is defined as

$$a_t = \sigma(W a_{t-1} + U x_t + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (\text{A.11})$$

selected for its probabilistic interpretation and contraction properties. Feature selection follows

$$S_{\Delta t} = \{i : \Delta H_t(i) > \tau_s(i)\}, \quad \phi(s) = W s + c, \quad (\text{A.12})$$

with thresholds applied either to $\max_i a_t(i) > \theta$ or to $\text{mean}(a_t) > \theta$. Symbolic injection into the RDF graph is defined as

$$\Psi_{\text{inject}}(a_t) = \{(v_i, : \text{hasActivation}, a_t(i)) \mid v_i \in V, a_t(i) > \theta\}. \quad (\text{A.13})$$

Stability analysis in discrete time leads to

$$L_{t+1} - L_t \leq -\epsilon \|a_t - \theta\|^2, \quad \epsilon > 0, \quad (\text{A.14})$$

showing Lyapunov decrease under gated updates. Computational complexity is parameterized by input size $n = |X_t|$: ARNN inference has cost $O(|E|)$ on sparse graphs, while RDF updates cost $O(r)$ for r SPARQL rules.

C. Modular Semantic Update

For each subgraph $G_t(i)$, coherence is defined as

$$\text{Coh}(G_t(i)) = 1 - \frac{\# \text{ inconsistent triples in } G_t(i)}{\# \text{ total triples in } G_t(i)}, \quad \kappa = 0.95. \quad (\text{A.15})$$

Graph partitioning can be formulated as a spectral clustering problem, with approximation quality supported by Cheeger bounds. Activation values may be aggregated (e.g., through means) to compute update probabilities, mitigating local noise:

$$P(\text{update}_i) = \sigma(\text{mean}(a_t(i)) + \Delta H_t(i) - \tau_s(i) - \theta(i)). \quad (\text{A.16})$$

The Lyapunov functional can be extended to incorporate inter-subgraph coupling:

$$L = \sum_i L_i + \lambda \sum_{i \neq j} |E_{ij}|, \quad (\text{A.17})$$

$$L_i = \alpha \|\Delta H_t(i)\|^2 + \beta (1 - \text{Coh}(G_t(i)))^2.$$

D. Neurosymbolic UI and Other Elements

For the neurosymbolic user interface, distance-preserving properties should be maintained,

$$\|V_{\text{interact}}(u) - V_{\text{interact}}(v)\| \approx d(u, v). \quad (\text{A.18})$$

Threat probability P_{threat} may be modeled as the stationary distribution of a Markov chain over graph states. In SPARQL propagation, exponential decay is preferred:

$$P_{\text{prop}} = a_t \cdot e^{-\rho d}, \quad \rho > 0, \quad (\text{A.19})$$

where d denotes graph distance. Algorithmic pseudocode should explicitly include weight updates:

$$W_{t+1} = W_t - \eta \nabla L_{\text{total}}, \quad (\text{A.20})$$

with η the learning rate.

E. General Integration

Key definitions such as W_{gt} , ΔH_t , and $\text{Coh}(G_t)$ should be integrated into the main body of the paper to ensure self-containment. Appendix subsections can then be used to provide concise proofs of convergence, Lipschitz continuity, and stability conditions. For multimodal Digital Twins, entropy fusion should be formalized as

$$H(X_t) = H(\text{physical}) + H(\text{virtual}) - I(\text{physical}; \text{virtual}), \quad (\text{A.21})$$

where I denotes mutual information, thus explicitly modeling synchronization mismatches between physical and virtual domains.

Moreover, this appendix consolidates the proposed improvements to the mathematical formalism of “ Ψ -Risk-DT: A Neurosymbolic Framework for Adaptive Zero-Day Threat Detection in Digital Twin Ecosystems”. Refinements focus on definitions, consistency, differentiability, convergence, and computational clarity.